

Label-free cold start for active learning beats a supervised evolutionary envelope

— if the envelope is honest: a circularity audit

Gilsiley Henrique Darú¹
Gustavo Valentim Loch¹

¹Graduate Program in Numerical Methods in Engineering (PPGMNE),
Federal University of Paraná (UFPR), Curitiba, Brazil

Draft v0.1 — July 18, 2026

Abstract

The first labeled set L_0 decides how an active learning loop starts — and in short-text classification with hundreds of imbalanced classes, it decides a lot: we measure a 6.4-point accuracy spread at $|L_0| = 100$ caused by the random draw alone (47 sizes $\times$ 30 repetitions, independently re-executed with ≤ 0.7 -point divergence). To estimate how much deliberate composition can yield, a genetic algorithm optimizes L_0 composition under four objectives, revealing a wide envelope (up to 25.9 points between best and worst individuals at $|L_0| = 100$) that narrows with size. We then show that DRI-SL — a deterministic, label-free, linear-cost heuristic combining semantic density (embedding-cluster representativeness) with lexical variety — *outperforms the best individual found by the supervised optimizer* at every size from 100 to 5,000 (e.g., 67.4% vs. 62.2% accuracy at $|L_0| = 1,000$). Finally, a methodological audit shows the standard protocol of reporting evolutionary envelopes on the same partition used as fitness inflates them — by 6.3 points of macro-F1 at $|L_0| = 500$ in our re-execution with an anti-circularity protocol — which means published evolutionary bounds for subset selection are optimistic and our comparison is conservative. Composition, not just quantity, governs the cold-start regime; it can be captured without any labels; and envelopes must be reported on untouched partitions.

Keywords: active learning; cold start; instance selection; genetic algorithms; evaluation methodology; short text.

1 Introduction

Pool-based active learning assumes a model good enough to guide selection — which presupposes an initial labeled set L_0 . In the cold-start regime the loop’s later efficiency is hostage to this first draw (Bayer and Reuter, 2024; Attenberg and Provost, 2010). Three questions structure this paper: (RQ1) how much does the *composition* of L_0 matter, at fixed size? (RQ2) how much headroom does deliberate composition have over random draws — an envelope we estimate with a supervised evolutionary optimizer? (RQ3) can a label-free constructive heuristic capture that headroom? A fourth, methodological question emerged from answering the second: (RQ4) are evolutionary envelopes, as usually reported, even honest?

We answer on a hard instance of the problem: 250k short product descriptions (4–50 characters, abbreviations) in Portuguese, 621 imbalanced classes, where class coverage — not instance count — dominates early performance.

2 Related work

Cold-start-aware AL uses pretrained representations to pick informative seeds: ALPS exploits language-model surprisal (Yuan et al., 2020); BERT-era studies adapt AL to fine-tuning (Ein-Dor et al., 2020; Griefhaber et al., 2020); DEUCE combines textual and class-prediction diversity (Guo et al., 2025); LLM-based selectors push the idea further (Bayer and Reuter, 2024). Density/cluster-based selection is classical (Settles and Craven, 2008; Nguyen and Smeulders, 2004; Dasgupta and Hsu, 2008), as are batch-diversity concerns (Hoi et al., 2006). Evolutionary algorithms (Holland, 1975; Goldberg, 1989) are widely used for instance and subset selection; we are not aware of prior work auditing the *evaluation circularity* of their reported envelopes in this setting — fitness measured and reported on the same partition — although the underlying overfitting-to-selection phenomenon is well known in model selection.

3 Setting

Classifier: PVBIn, a per-class prototype model over binary term vectors, cheap enough for tens of thousands of retrainings (Darú, 2024). Data: deduplicated short product descriptions, 621 classes, heavy long tail; evaluation on a fixed untouched test set (20,000 instances in the re-execution protocol). Metrics: global accuracy and macro-F1, the latter primary under imbalance (Sokolova and Lapalme, 2009).

4 RQ1: sensitivity to the draw

Across 47 sizes (10 to 200,000) with 30 random repetitions each: mean accuracy grows from 6.7% ($|L_0| = 10$) through 24.7% (100), 55.9% (1,000) and 76.9% (10,000), saturating near 89.1% (200,000); macro-F1 lags (0.4% $\rightarrow$ 3.4% $\rightarrow$ 17.8% $\rightarrow$ 43.3% $\rightarrow$ 70.4%), reflecting the cost of covering minority classes. At $|L_0| = 100$, accuracy across repetitions spans **21.7–28.1%** — a **6.4-point** spread caused exclusively by the draw. The mechanism is coverage: at $|L_0| = 10$ a typical draw covers 9 of 621 classes; at 1,000, about 255; macro-F1 tracks the coverage curve closely. Good and bad L_0 of the same size *exist*; the question becomes whether they can be found without labels.

Independent re-execution. The full study was re-executed in a clean-room reimplementaion with a stricter protocol (deduplication *before* partitioning; fixed test set), on a reduced grid (15 log-spaced sizes $\times$ 10 repetitions). Agreement is within 0.7 points at every size, always in the expected direction (the deduplicated protocol is slightly harder), with implementation fidelity additionally verified by exact equality of score matrices.

5 RQ2: the evolutionary envelope

A genetic algorithm over subset membership (100 generations; the full configuration ships in the released artifacts) optimizes L_0 composition for four objectives (max/min $\times$ accuracy/macro-F1) at ten sizes. The envelope is wide where it matters: at $|L_0| = 100$, best-minus-worst final individuals span **25.9 accuracy points**; evolutionary gain over the first generation (e.g., +10.1 points at 100, +11.7 at 50) confirms the optimizer exploits real structure, not noise. The envelope narrows monotonically with size (at 30,000: +0.8), consistent with RQ1: composition is a small-regime lever.

6 RQ3: a label-free heuristic vs. the supervised optimizer

DRI-SL (Density-Representative Instance Selection with Lexical variety) is deterministic given the embedding model and uses no labels: embed the pool with dense text embeddings, cluster (k -means), allocate the budget proportionally to cluster mass, and fill each cluster’s quota greedily preferring instances that add unseen lexical tokens — linear cost after clustering. Table 1 compares it with the GA envelope.

Table 1: Cold start: DRI-SL vs. GA envelope (final generation), PVBIn. GA values inherit the original protocol (fitness partition = report partition) and are therefore optimistic — see Section 7.

$ L_0 $	DRI-SL		GA (best)		GA (worst)	
	Acc	M-F1	Acc	M-F1	Acc	M-F1
100	41.2	6.9	38.8	6.5	5.8	1.8
500	59.2	18.5	56.0	18.0	36.2	6.5
1,000	67.4	25.8	62.2	24.7	49.3	12.6
2,500	73.4	36.5	69.3	32.6	63.9	24.1
5,000	76.9	44.1	74.1	40.1	70.8	31.9

A constructive heuristic with zero label access beats an optimizer that consumed thousands of supervised evaluations, at every size — strong evidence that semantic density plus lexical variety capture precisely the structure that makes an L_0 informative (class coverage through cluster coverage; feature coverage through lexical novelty). At 100 the margin over the GA’s best is 2.5 points; over the GA’s *worst*, 35 points — the cost of a bad draw that DRI-SL eliminates deterministically.

7 RQ4: the circularity audit

The GA numbers above inherit the field’s standard protocol: fitness evaluated on a partition, and the *same* partition reported as the result. Our re-execution applies an anti-circularity protocol — fitness on a selection partition, final individuals *re-evaluated once* on the untouched test set. The mechanism reproduces (best individual gains +5.2 accuracy points at $|L_0| = 50$), but the honest re-evaluation reveals the inflation: for macro-F1 maximization at $|L_0| = 500$, final fitness reads 19.4% while the same individual measures **13.1%** on untouched data — **6.3 points of overfitting to the fitness partition** (for accuracy, ≈ 1 point). Two consequences: (i) published

evolutionary envelopes for subset selection should be read as optimistic bounds unless the protocol states otherwise; (ii) our central comparison is *conservative* — DRI-SL beats an inflated opponent.

8 Conclusion

Composition of the initial labeled set is a first-order factor in cold-start active learning for long-tailed short text (6.4-point spread from the draw alone at $|L_0| = 100$); a supervised evolutionary optimizer confirms wide headroom; a deterministic label-free heuristic captures that headroom, beating the optimizer’s best individual up to $|L_0| = 5,000$; and evolutionary envelopes must be reported on untouched partitions — ours was inflated by up to 6.3 macro-F1 points before correction. In the continuous (post-cold-start) regime, companion work shows the lexical-variety component should be dropped in favor of plain prediction-stratified sampling; the two results together delimit where each ingredient earns its keep. All artifacts are released.

Artifacts. github.com/GHDaru/activelearning (experiments/p1, p2).

References

- Josh Attenberg and Foster Provost. Why label when you can search?: Alternatives to active learning for applying human resources to build classification models under extreme class imbalance. *Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 423–432, 2010. doi: 10.1145/1835804.1835859.
- Manuel Bayer and Christoph Reuter. Activellm: Large language model-based active learning for textual few-shot scenarios. arXiv preprint arXiv:2405.10808, 2024. URL <https://arxiv.org/abs/2405.10808>.
- Gilsiley Henrique Darú. Categorização de produtos em e-commerce: avaliação do método argmax para classificação de descrições curtas em português. Master’s thesis, Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, 2024.
- Sanjoy Dasgupta and Daniel Hsu. Hierarchical sampling for active learning. In *Proceedings of the 25th International Conference on Machine Learning (ICML ’08)*, pages 208–215. ACM, 2008.
- Lior Ein-Dor, Alon Halfon, Avi Gera, Ehud Shnarch, Leonard E. Dankin, Liat Choshen, Matan Danilevsky, Ranit Aharonov, Yoav Katz, and Nadav Slonim. Active learning for bert: An empirical study. In *EMNLP 2020*, pages 7949–7962, 2020.
- David E. Goldberg. *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison-Wesley, 1989.
- Daniel Grieshaber, Julia Maucher, and Ngoc Thang Vu. Fine-tuning bert for low-resource natural language understanding via active learning. In *COLING 2020*, pages 1158–1171, 2020.
- Jiaxin Guo, C. L. Philip Chen, Shuzhen Li, and Tong Zhang. DEUCE: Dual-diversity enhancement and uncertainty-awareness for cold-start active learning. *Transactions of the Association for Computational Linguistics*, 12:1736–1754, 2025. doi: 10.1162/tacl_a_00731.

- Steven C. H. Hoi, Rong Jin, and Michael R. Lyu. Large-scale text categorization by batch mode active learning. In *15th International Conference on World Wide Web*, pages 633–642, 2006.
- John H. Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, 1975.
- Hieu T. Nguyen and Arnold W. M. Smeulders. Active learning using pre-clustering. In *Proceedings of the 21st International Conference on Machine learning (ICML '04)*, page 79. ACM, 2004.
- Burr Settles and Mark Craven. An analysis of active learning strategies for sequence labeling tasks. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing (EMNLP '08)*, pages 1070–1079. Association for Computational Linguistics, 2008.
- Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009.
- Michelle Yuan, Hsuan-Tien Lin, and Jordan Boyd-Graber. Cold-start active learning through self-supervised language modeling. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7935–7948, Online, 2020. Association for Computational Linguistics.